

10. Multi-tenant Retrieval and Access Control

In a multi-tenant RAG system, each document, chunk, embedding, and retrieval request should be associated with a tenant identifier. Client isolation should happen before the LLM receives any context.

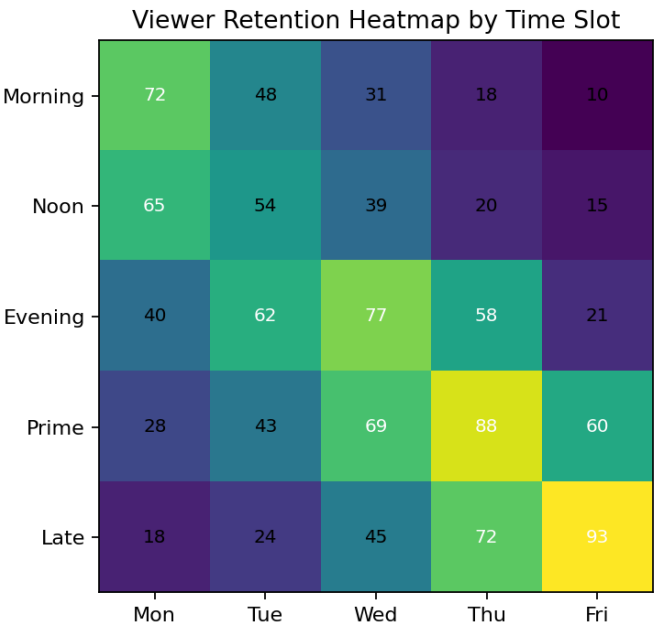
Layer	Control	Failure mode if missing
Authentication	Identify user and organization	Unknown user may access application
Authorization	Map user to client_id and role	User may retrieve another clients documents
Vector retrieval	Filter by client_id or namespace	Retriever may return unauthorized chunks
Prompt construction	Send only allowed context	LLM may see sensitive data
Audit logging	Record query, source chunks, user_id	No traceability during incident review

Correct flow: authenticate user, get client_id from backend, retrieve only matching chunks, build prompt with authorized context, return answer with citations.

Wrong flow: retrieve from all clients and tell the model to ignore unauthorized content. This is unsafe because the LLM should not be used as the access-control boundary.

11. Analytics Image: Retention Heatmap

The heatmap below simulates viewership retention by time slot. A parser that ignores chart images will miss useful business context. OCR can extract axis labels and numbers, but chart understanding may require a vision model.



Observation	Supporting value	Scheduling implication
Prime Friday performs best	93 retention score	Place premium content in Friday late/prime slots
Early week late slots underperform	10 to 21 score range	Avoid launching new series in weak slots
Evening improves midweek	77 on Wednesday evening	Use midweek evening for discovery content

For RAG, chart captions should be indexed. If chart values are critical, extract them into structured metadata or a companion table.

12. Evaluation Dataset

A RAG system should be evaluated separately for retrieval quality and answer quality. This page includes a miniature evaluation set with expected source references. It is useful for testing whether citations point to the correct section.

Test ID	Question	Expected source	Expected answer element	Failure signal
Q-001	What is the refund window?	Policy Rules table	7 days	Answer says 30 days
Q-002	Which clause covers data residency?	Clause 9.1	Approved processing region	No clause citation
Q-003	What is P1 API response target?	SLA table	15 minutes	Wrong severity row
Q-004	Which documents belong to BLR-ACME-2026?	Relationship table	Five related documents	Only one document returned
Q-005	Who owns access review?	Policy Rules table	Security	Owner missing

Evaluation should include adversarial questions, unrelated questions, and questions that require multi-hop retrieval across related documents.

13. Edge Cases for Parsers

The following cases often break real document ingestion pipelines. They are included here as guidance for testing parser quality before moving to embeddings and vector storage.

Edge case	Example	Recommended handling
Repeated headers and footers	Page number, confidentiality banner	Remove or store separately as metadata
Hyphenated line breaks	termi- nation assistance	Normalize during cleaning
Rotated tables	Landscape appendix in PDF	Use layout-aware parser or OCR
Merged cells	Pricing table with grouped plans	Preserve hierarchy in row text
Scanned signatures	Signature block as image	OCR if text is needed; store image reference
Boilerplate navigation	Website header and footer	Use boilerplate removal
Duplicate chunks	Same policy in FAQ and PDF	Deduplicate using source and hash

Bad parsing creates bad chunks. Bad chunks create bad retrieval. Bad retrieval creates bad answers.

15. Final Ingestion Checklist

Use this checklist before sending parsed content into chunking and embeddings. It helps identify whether the data is ready for production RAG.

Checklist item	Status to verify	Why it matters
Text extraction	Paragraphs are readable and ordered	Prevents noisy chunks
Table extraction	Rows, columns, headers, and footnotes preserved	Protects factual answers
Image handling	Captions indexed and OCR done if required	Avoids missing visual information
Metadata	source, page, client_id, document_id captured	Enables citations and access control
Chunking	Chunks preserve meaning and section boundaries	Improves retrieval relevance
Access control	Retrieval filters use authenticated client_id	Prevents data leakage
Evaluation	Golden questions tested with citations	Measures real answer quality

Summary: RAG is about knowledge access. Fine-tuning is about behavior adaptation. For document intelligence, parsing quality is the foundation. If extraction is weak, no embedding model or LLM can fully fix the missing context.

End of synthetic 15-page parsing test document.

Appendix A: Complex Clause Responsibility Matrix

Grouped clauses with owner/backup split inside the same responsibility cell. This page is useful for testing row grouping, merged-looking labels, split responsibility cells, and long evidence text.

Parsing challenge: preserve row boundaries, nested headers, split cells, grouped labels, numeric values, and footnotes/context around the table.

and a split responsibility cell where owner and backup are shown inside the same row.

Clause Group	Obligation	Responsible Team		Trigger	Evidence Required	Risk
Data Protection	Delete client data after contract termination unless retention is legally required.	Owner	Compliance	Termination notice received	Deletion certificate + audit log export	High
		Backup	Legal			
	Notify client about any confirmed data incident within 72 hours.	Owner	Security	Incident classified as confirmed breach	Incident report, timeline, notification proof	Critical
		Backup	DPO			
Billing	Apply annual platform fee adjustment only after renewal confirmation.	Owner	Finance	Renewal order approved	Approved renewal sheet + invoice draft	Medium
		Backup	CSM			
	Do not bill inactive campuses during suspension period.	Owner	Revenue Ops	Campus status = suspended	ERP campus status export	High
		Backup	Finance			
Support	Provide P1 response within 30 minutes during school operating hours.	Owner	Support L2	Ticket priority = P1	Ticket timestamps + agent assignment log	High
		Backup	Ops Manager			
	Escalate unresolved P2 tickets after 4 business hours.	Owner	Support L1	Ticket age > 4 business hours	Escalation log	Medium
		Backup	Support L2			

Table 1: Added as complex parsing appendix for table extraction, OCR fallback, and layout-aware RAG testing.

Appendix B: Regional Pricing and Usage Add-on Matrix

Pricing table with multi-level headers, regional columns, add-on columns, billing rules, exception rows, and mixed numeric/text values.

Parsing challenge: preserve row boundaries, nested headers, split cells, grouped labels, numeric values, and footnotes/context around the table.

Plan	Student Volume	Annual Platform Charges		Usage Add-ons		Billing Rule	
		India Region		International	SMS		WhatsApp
Starter	0 - 2,000 students	Base	INR 4.5L	USD 7,200	INR 0.18/message	INR 0.42/message	Quarterly advance
		Support	INR 60K				
Growth	2,001 - 10,000 students	Base	INR 11L	USD 18,000	INR 0.15/message	INR 0.38/message	50% advance + monthly usage
		Support	INR 1.4L				
Enterprise	10,001+ students	Base	Custom	Custom	Negotiated	Negotiated	Signed order form required
		Support	Included				
Exception	Government schools	Discount may apply after approval			No discount on pass-through cost		Requires CFO approval

Table 2: Added as complex parsing appendix for table extraction, OCR fallback, and layout-aware RAG testing.

Appendix C: Invoice Line Items with Tax Split

Invoice-style line item table with item groups, quantity, rate, CGST/SGST split, totals, and summary row. Useful for invoice parsing and tax extraction tests.

Parsing challenge: preserve row boundaries, nested headers, split cells, grouped labels, numeric values, and footnotes/context around the table.

Item Group	Line Item	Qty	Rate	Tax Split		Total
				CGST	SGST	
ERP Platform	Annual School360 Enterprise Subscription - 12 campuses	1	INR 11,00,000	9%	9%	INR 12,98,000
	Parent communication add-on - estimated message pack	2,00,000 msgs	INR 0.38/msg	9%	9%	INR 89,680
Implementation	Data migration + training + go-live support	1	INR 2,40,000	9%	9%	INR 2,83,200
Summary	Subtotal and taxes	-	INR 14,16,000	INR 1,27,440	INR 1,27,440	INR 16,70,880

Table 3: Added as complex parsing appendix for table extraction, OCR fallback, and layout-aware RAG testing.

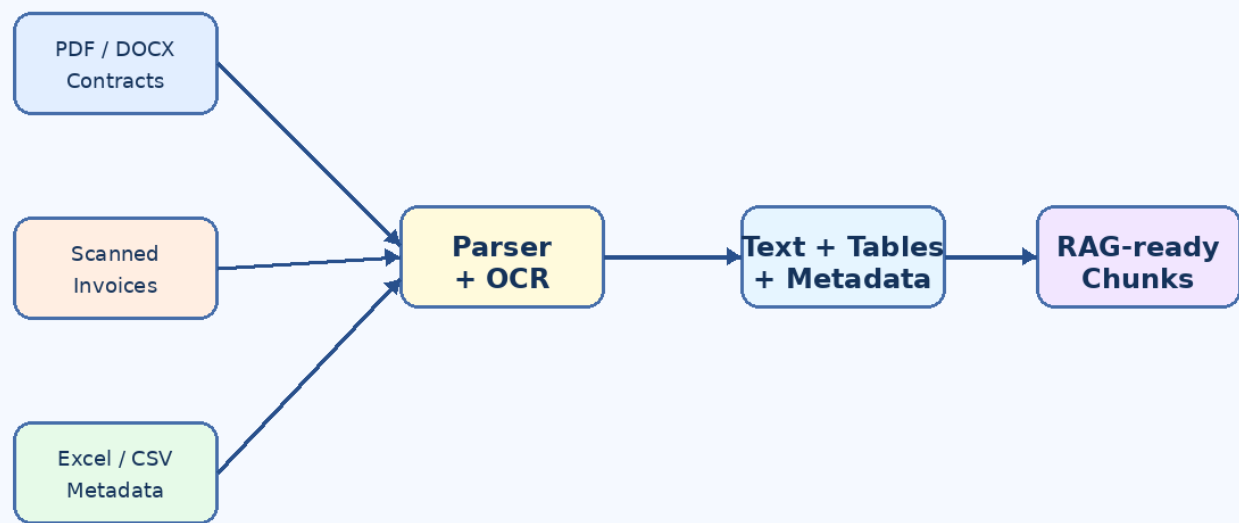
Appendix D: Multimodal Document Processing Flow

Diagram-style image showing how PDFs, DOCX files, scanned invoices, Excel/CSV metadata, parser/OCR, extracted text/tables/metadata, and RAG-ready chunks connect.

Parsing challenge: image text, arrows, labels, and captions may not appear in normal PDF text extraction. OCR or multimodal parsing may be required.

Visual Appendix 1: Multimodal Document Flow

Parsing challenge: preserve text, tables, images, OCR output, and metadata before chunking.



Expected extraction: content blocks with source file, page, section, modality, and confidence score.

Added at the end for complex image parsing, OCR fallback, layout-aware extraction, and multimodal RAG testing.

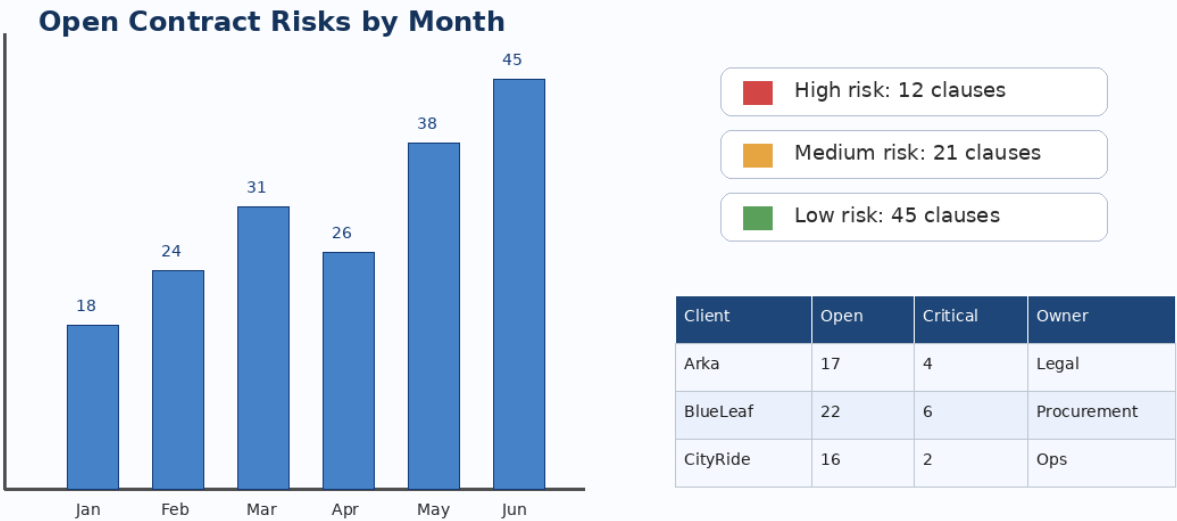
Appendix E: Contract Risk Dashboard Image

Dashboard-style image containing a bar chart, legend, and small matrix. Useful for testing chart extraction, numeric value capture, and caption-aware indexing.

Parsing challenge: extract chart title, bar values, legend labels, and table values from an embedded image.

Visual Appendix 2: Contract Risk Dashboard Snapshot

Parsing challenge: extract chart labels, legends, values, and nearby explanatory text.



Expected extraction: chart title, series values, legend labels, table values, and risk summary.

Added at the end for complex image parsing, OCR fallback, layout-aware extraction, and multimodal RAG testing.